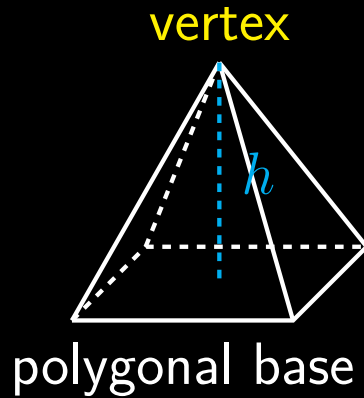


Derivation of the Volume Formula for Square and Rectangular Pyramids

Jonathan R. Bacolod

A **pyramid** is a solid figure with one **polygonal base** and triangular faces meeting at a common **vertex**.

Parts of a Pyramid

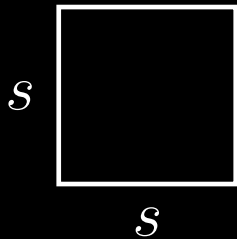


$h = \text{height}$

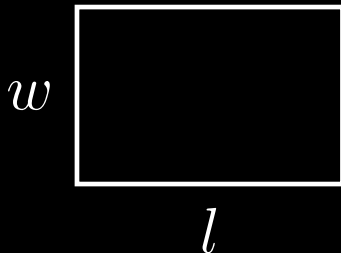
Square and Rectangular Pyramids

A **square pyramid** has a square base, while a **rectangular pyramid** has a rectangular base.

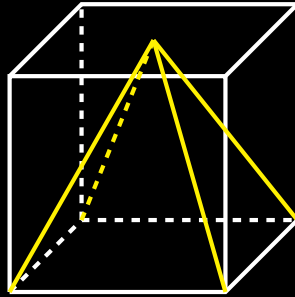
square base



rectangular base



Same Base, Same Height



A pyramid having the same base and height as a rectangular prism occupies exactly **one-third** of the prism's volume.

One-Third of a Prism

The **volume of a pyramid** is one-third of the volume of a prism with the same base and height.

$$V_{\text{pyramid}} = \frac{1}{3} \times V_{\text{prism}}$$

Think: 3 pyramids fill 1 prism.

Volume Formula

The **formula** for the volume of a pyramid is

$$V = \frac{1}{3}Bh$$

where B is the **area of the base** and h is the **height**.

Square Pyramid

For a square pyramid, the base area is $B = s^2$.

So

$$V = \frac{1}{3}s^2h$$

where s is the side length of the square base.

Rectangular Pyramid

For a rectangular pyramid, the base area is $B = lw$. So

$$V = \frac{1}{3}lwh$$

where l is the length, w is the width, and h is the height.

Summary of Formulas

Pyramid	Base Area B	Volume V
Any pyramid	B	$\frac{1}{3}Bh$
Square	s^2	$\frac{1}{3}s^2h$
Rectangular	lw	$\frac{1}{3}lwh$

Steps to Find the Volume

1. Find the base area B (if it is not given).
2. Identify the height h .
3. Substitute into $V = \frac{1}{3}Bh$.
4. Compute and write the **cubic units**.

Examples

Example 1

A square pyramid has a base with side length 5 cm and a height of 12 cm. Find its volume.

$$\begin{aligned} V &= \frac{1}{3}s^2h \\ &= \frac{1}{3}(5)^2(12) \\ &= \frac{1}{3}(300) \\ &= 100 \text{ cm}^3 \end{aligned}$$

Example 2

A square pyramid has a base area of 64 cm^2 and a height of 9 cm. Find its volume.

$$\begin{aligned} V &= \frac{1}{3}Bh \\ &= \frac{1}{3}(64)(9) \\ &= \frac{1}{3}(576) \\ &= 192 \text{ cm}^3 \end{aligned}$$

Example 3

A rectangular pyramid has a base measuring 9 cm by 6 cm and a height of 10 cm. Find its volume.

$$\begin{aligned} V &= \frac{1}{3}lwh \\ &= \frac{1}{3}(9)(6)(10) \\ &= \frac{1}{3}(540) \\ &= 180 \text{ cm}^3 \end{aligned}$$

Example 4

A pyramid has a rectangular base with area 120 cm^2 and a height of 6 cm . Find its volume.

$$\begin{aligned} V &= \frac{1}{3}Bh \\ &= \frac{1}{3}(120)(6) \\ &= \frac{1}{3}(720) \\ &= 240 \text{ cm}^3 \end{aligned}$$

Example 5

A prism has a volume of 300 cm^3 . A pyramid has the same base and height as the prism. Find the volume of the pyramid.

$$\begin{aligned} V_{\text{pyramid}} &= \frac{1}{3} \times V_{\text{prism}} \\ &= \frac{1}{3}(300) \\ &= 100 \text{ cm}^3 \end{aligned}$$

Practice Exercises

Practice 1

Find the volume of the pyramid.

A square pyramid has a base with side length 6 cm and a height of 9 cm.

Practice 2

Find the volume of the pyramid.

A square pyramid has a base area of 81 cm^2 and a height of 15 cm .

Practice 3

Find the volume of the pyramid.

A rectangular pyramid has a base measuring 10 cm by 8 cm and a height of 9 cm.

Practice 4

Find the volume of the pyramid.

A square pyramid has side length 12 m and height 10 m.

Practice 5

Find the volume of the pyramid.

A rectangular pyramid has dimensions 14 cm by 9 cm with height 12 cm.

Practice 6

Find the volume of the pyramid.

A pyramid has a rectangular base with area 96 cm^2 and height 15 cm .

Activities

Activity 1

Find the volume of the pyramid.

A square pyramid has a base with side length 8 cm and a height of 12 cm.

Activity 2

Find the volume of the pyramid.

A square pyramid has a base area of 100 cm^2 and a height of 18 cm .

Activity 3

Find the volume of the pyramid.

A prism has a volume of 720 cm^3 . A pyramid has the same base and height as the prism.

Activity 4

Find the volume of the pyramid.

A square pyramid has side length 15 m and height 8 m.

Activity 5

Find the volume of the pyramid.

A rectangular pyramid has dimensions 18 cm by 10 cm with height 9 cm.

Activity 6

Find the volume of the pyramid.

A pyramid has a rectangular base with area 150 cm^2 and height 14 cm .

Thank you!